Modern Staybolt Inspection Methods*

Device obviates necessity of removing caps from flexible staybolts for periodic inspection

By E. S. Fitz Simmons†

THE development of the flexible staybolt in this country dates back to about 1890, and early experimentation was stimulated by the enormous number of breakages which occurred in the rigid type of staybolt. Probably the first type of flexible staybolt which could be readily inspected without removal was one of the round headed design, using a sleeve and removable cap, which was designed by John B. Tate of the Pennsylvania Railroad and introduced by the Flannery Bolt Company about 1904.

This design gave promise of eliminating many of the weaknesses of former types and provided a means, though an expensive one, of ascertaining the condition of the bolt without removing it from the boiler. For these reasons it was quite generally tried out by many railroads and the results obtained were so satisfactory that its adoption and general use almost immediately followed.

In 1905 a committee of the American Railway Association reporting on flexible staybolts said of it: "Though of comparatively recent date, it has been received with great favor, indicating the rapidity with which a new bolt of promising design is given a practical trial in the effort to cure the staybolt trouble." Further in this same report the committee stated the following: "A simple means of positively detecting cracked and broken bolts is an improvement needed by all flexible bolts that are in service at the present time. The removal of the cap, which will allow an examination of the bolt, is the only absolute means possible with flexible bolts of the present design."

In 1911 the act commonly known as the Locomotive Boiler Inspection Law became effective, under authority of which, rules requiring among other things the removal of caps from flexible staybolts for inspection purposes at special stated intervals, were promulgated and enforced.

The generally satisfactory service rendered and the very small percentage of breakage to total number in service (this type having become practically standard) created a doubt in the minds of many railroad mechanical officers of the justification of such a rule or requirement, and it was quite generally, and for a time quite strenuously, opposed. However, the Locomotive Inspection Bureau, through its corps of inspectors, had indisputable evidence of the necessity for thorough and regular inspection, for while it was admitted that the percentage of breakage to the total number in service was small, it was shown beyond question of doubt that in certain instances breakage did occur to such an extent as to become a menace to life and property.

Investigations were begun to determine the cause of breakage in an endeavor to apply a remedy. A number of investigations confirmed the absolute necessity for careful and frequent inspections and also clearly demonstrated the need of a more accurate and positive method of test than the mere removal of the caps and an examination of the bolt heads.

Instances were found where, under bad water conditions, accumulations of scale between the body of the bolt and the inner wall of the sleeve were such as to solidly lock the bolt in the sleeve. The result was to render ineffectual the flexible feature and to cause the bolt to break at the firebox sheet.

During the past ten years, and more diligently during the past five years, after these abnormal conditions had been prominently developed, constant effort has been made to perfect a method of inspection that would provide greater safety and insure against loss of life, injury to persons and damage to property.

The first actual service test of such a method was made in August, 1920, on large Mallet type locomotives where abnormal conditions existed and where breakage of flexible bolts was excessive. This method consisted of the installation of flexible bolts with telltale holes extending from the firebox end entirely through the body section and terminating within the bolt head, and the testing of these bolts at regular intervals with a specially constructed instrument to determine definitely that the telltale holes were open and, therefore, operative throughout their entire length. This was accomplished by so constructing the instrument that an electrical circuit would be established when contact was made with the end of the telltale hole.

During this service test, two distinct features were developed.

First—that moisture, due to temperature changes, condensed in the telltale hole and formed rust or iron oxide which gradually increased to such an extent as to interfere with the insertion of the testing instrument. In seeking a remedy for this condition, various methods were tried, the most satisfactory and the one at present in use being the electro-plating of the walls of the telltale hole with copper. Four years experimental work and service tests have demonstrated the complete success of eliminating the difficulty by this method.

Second—that cinder and other foreign matter accumulated in the telltale holes to such an extent as to require excessive labor to dislodge and remove it in order to permit the insertion of the testing instrument. The cost and time required for this made the method impractical. After numerous experiments it was found that a closure of fire-proof porous material applied in the end of the telltale hole prevented the accumulation and at the same time permitted leakage of steam or water in case of fracture, and that it was also readily and cheaply removable to allow insertion of the testing instrument.

Application of Telltale Flexible Bolts

The telltale flexible bolt is identically the same as the Tate bolt, with the addition of a telltale hole extending through the entire length of the body section and terminating within the head of the bolt. The walls of the telltale hole are copper-plated to prevent rust or corrosion within the hole and to prevent them from becoming closed from this cause.

They are applied exactly the same as the ordinary flexible bolt. If the method of riveting closes the end of the hole, it may easily and quickly be re-opened, after which a porous fireproof closure is applied that will prevent the accumulation of foreign matter from entering the telltale hole and that will permit leakage of steam or water in case of a break or fracture, which serves as a daily indicator of the condition of the bolt.

Inspection of Telltale Flexible Staybolts

In addition to depending on the leakage through the telltale hole, an inexpensive method of periodically checking

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out some of the features of the construction of a boiler after this information has been furnished.

Fireholes.—Fireholes are made in various types and sizes depending on the type the railroad company uses or prefers. Some mechanical stokers also enter into the design and location of the firehole. The most simple and used design of firehole is shown on boiler drawing Fig. 1. The connection between the backhead and the firebox back sheet is made by flanging the backhead and the firebox sheet into the water leg. These two flanges are then connected by a simple riveted lap seam with ¾-inch diameter, patched bolts, 2-inch pitch and countersunk heads or welded with a butt joint or ½-inch lap welded joint. A ¾-inch radius is used for the backhead flange. The firebox sheet flange radius should be more than ¾ inch when the width of the water leg will permit a larger

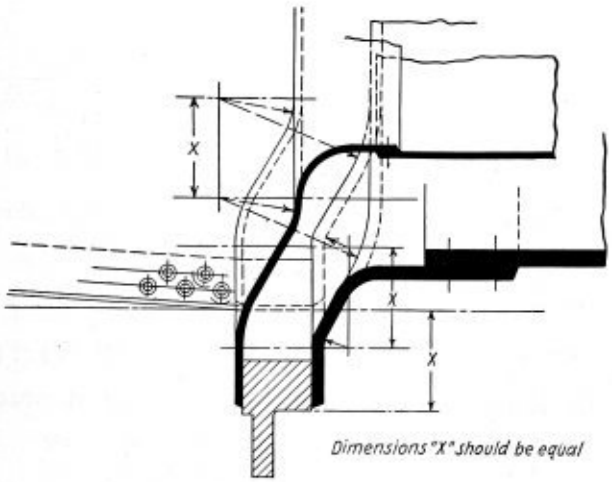


Fig. 47.—Sloping bend construction of throat

one to be used, which is most always possible with the welded type of flange. The larger radius serves to relieve the sheet from cracking or springing a leak at the connection.

When the firedoor is opened or closed the firebox sheet is subject to considerable expansion and contraction. This trouble is eliminated when the boiler is operated with a mechanical stoker. However, the large flange radius should be used for various causes of expansion and contraction.

The firebox and the outside shell are riveted to a solid bar at the bottom called the firebox ring. This ring is usually made with four pieces welded together—the sides are forged iron and the ends cast steel. The ring is 3½ inches to 4 inches thick, double riveted with ⅞-inch swell neck, through rivets about 3 inches pitch and 1¼ inches between the rivet lines. The ring should be ¼ inch below the bottom edge of the sheets. The front and back edge of the firebox ring at the back end and at the throat should be parallel, for application of the ring in the firebox.

The firebox and the outside shell above the firebox ring are stayed together to resist the steam pressure. The stays in the water legs are called "staybolts." The stays supporting the crown sheet to the roof sheet are called "radial stays" for reason of being radial to the crown sheet. The roof sheet is self-supporting because of its circular shape. The extent of the radial stays is at some point on the crown corner radius where the space between the inside surface of the crown sheet and the outside surface of the roof sheet is not less than 8 inches.

Radial stays have upset ends and when the distance over sheets is less than 8 inches it is not practical to make the upsets. As a rule all radial stays are rigid and have straight threads in the roof sheet end. From six to ten center longitudinal rows of radial stays have taper threaded ends in the crown sheet end. The top of the crown being subject to more intense heat, the taper end type of stay supports the sheet and keeps tight better than the straight end stay. The center rows or taper end stays are also referred to as "crown stays" and the straight threaded stays, or those outside of the taper end stays only, as "radial stays."

Rigid staybolts are made of round iron, threaded full length and vary in diameter from ⅞ inch to 1 inch, 1⅛ inches in special cases. All rigid staybolts and radial stays are headed over on the firebox and outside shell sheets. The head is low and slightly cone shaped. The stays should be made of hollow iron or have tell-tale holes about 3/16 inch diameter, drilled in the outer end to a depth beyond the inner surface of the shell sheet.

The above information relative to the kind of heads on the firebox end of crown stays refers to steel crown sheets only.

Crown stays for copper crown sheets have button heads or nuts on the crown sheet end.

Expansion Stays.—From two to four transverse rows of flexible stays are used at the front of the crown. The stays are called "expansion stays" and they relieve the firebox from severe strain—most of which occurs at the top-front of the firebox.

The calculation for the roof sheet thickness is based on the efficiency of the plate for rigid crown stays. The sleeve for expansion stays requires a larger hole in the sheet than rigid stays, therefore the efficiency of the roof sheet through the rows of expansion stays is reduced. This deficiency is corrected by the use of a liner on the inside of the roof sheet, when more than two transverse rows of expansion stays are used. The number of transverse rows of expansion stays used is based on the diameter of the back-boiler course. The longitudinal rows are usually carried around as far as the last radial stay row.

Location of Flexible Staybolts.—The staybolts in the firebox subject to the most strain should be flexible staybolts. A good sample for the location of flexible staybolts is shown by the two boiler drawings Figs. 1, 2, 9 and 10. The flexible staybolts are located in what is called the breaking zone and to describe this zone more clearly the staybolts should be located about as follows: Firebox sides—all the staybolts in the first three or four top rows—two rows up—front and back—beginning at the second row from the bottom and angled off to the top rows about as shown on Fig. 1. Backhead: Two outside rows—bottom row excepted. Throat: All staybolts except bottom row. Combustion chamber: The latest practice is to have all staybolts flexible instead of the six top rows as shown by the combustion chamber sections Fig. 10.

Type of Flexible Staybolt.—The type of flexible staybolt most commonly used is designed in three pieces, viz., bolt, sleeve and cap. The sleeves are of two types, viz., threaded and welded sleeves. The first type of sleeve requires the outside firebox or shell to be threaded for receiving the sleeves and the latter a hole through the shell without threads. The welding is around the sleeve and to the outer surface of the shell.

It is necessary to make a taper end flexible stay in four pieces, viz., bolt, ball nut, sleeve and cap. Taper end flexible stays are not recommended, having one more piece than the straight end flexible stay the manufactured cost is greater and it is not believed the taper end on a flexible stay adds much value to supporting the crown sheet. However, the same number rows of expansion stays as the taper end crown stays are made taper end when requested.

The cap for a welded sleeve, flexible staybolt projects about 1¼ inches beyond the outer surface of the shell, when the sleeve is approximately 90 degrees to the shell. A threaded sleeve cap projects about 1¾ inches beyond the shell, when the sleeve is at a right angle to the shell. A flush sleeve requires a larger hole in the shell than any other kind of sleeve.

(*To be continued*)

When possible, the liners should be so located that it will not be necessary to notch out the liner to clear the radial stays. It will also be found quite difficult in some cases, to obtain sufficient metal between two holes. The preferred minimum bridge distance for such cases is 1¼ inches, but may be less for small holes such as stays, studs and rivets.

LONGITUDINAL BRACE FEET

The longitudinal-brace-feet rivets should be located so that the brace pin will be approximately central between the stays and off the longitudinal line of stays a distance sufficient to prevent the stays from striking the foot. This is determined by laying-in the brace feet on the boiler cross-section.

WASHOUT PLUGS

The washout plugs are located centrally between the radial stay spaces, as shown on the boiler drawing, and the distance from the longitudinal row of stays as given on the boiler cross-section. When this distance is not the same, front and back, the front dimension is measured at the front row of stays and the back dimension at the back row of stays and the center line for the plugs is drawn through from these points.

STAYBOLTS

The staybolts are not laid-in on the roof side sheet at a longitudinal right angle to the crown side sheet, but are carried straight through in the same relative position from the front of roof and crown side sheets. The location vertically is at right angles to the straight part of the crown side sheet and radially for the curved part of the crown sheet. Therefore, the longitudinal location of the staybolts on the roof sheet are the same as on the crown sheet, except when set off the line to apply washout plugs, blow-off cocks, etc. The vertical locations of all the staybolts are not the same as on the crown sheet because the crown and roof sheets in the cross-section are not parallel to each other, above the bend line. Below the bend line the locations of staybolts are the same for both sheets. The two bottom rows of staybolts are made continuous on both the crown sheet and the roof sheet to simplify the method of dimensioning the staybolts on the roof sheet.

The side bend line of the boiler comes between these rows and by referring to the boiler cross-section, it will be seen that the space between the staybolts on the roof sheet is greater than on the crown sheet. This space, as shown by the boiler cross-section, is not exactly correct because the location has been projected from the longitudinal view of the boiler which is not the true relative location of the staybolts below the bend line though the location is correct from the last radial stay and according to the dimensions, front and back, as given on the boiler cross-section.

To determine the front and back dimension for this bottom space, refer to the diagram—Fig. 3—and take the difference between the distance from bottom of the firebox ring to the staybolt and bend line, which is ⅞ inch for this boiler. Lay-in this staybolt on the diagram cross-section—Fig. 4—⅞ inch below the bend line, front and back as shown. Refer to the crown-and-sides drawing—Fig. 6—and calculate the staybolt space, front and back, which is found to equal 55 15/16 inches ÷ 14, or 4 inches front, and 44⅞ inches ÷ 11, or 4.08 inches back. Lay-in these spaces on the diagram—Fig. 4—from the bottom staybolt, drawing the staybolt line at a right angle (radial if in the bend radius) to the side line then scale to the nearest 1/16 inch the space between the staybolts at the roof-sheet side line.

The staybolt spaces above the first bottom space will have the same spacing as the crown sheet, up to the crown corner radius, at which point they are radial to the corner radius. Those staybolts that come within the crown corner radius, and also the one which comes on the straight part of the sheet just below the tangent point of the crown radius and straight side line should be laid-in on the diagram—Fig. 4. Scale the distance between these stays at the roof neutral line.

Total all the dimensions for the radial-stay spaces and those for the staybolt spaces. These, plus U and U^1, should equal the length of the developing line from the top center to the bottom of the firebox ring, front and back. Measure the spaces on the developing lines.

The front and back row of staybolts are to be projected at 90 degrees from the developing lines and the lines drawn through from these points, as has been explained for the crown. The radial-stay lines should be drawn direct through from the developing lines.

The necessary longitudinal dimensions for locating the staybolts in the odd spaces at the back and lower right hand corner are scaled from the crown-sheet layout, and the dimensions scaled are used for the roof-sheet location. The dimensions are scaled at a right angle to the vertical staybolt line, but to simplify the method of dimensioning, and also because the difference between the straight and the sloped dimension is negligible, they are shown dimensioned along the staybolt line.

It will be noticed that the back line of staybolts on the roof sides is straight as is also the back line on the crown sides, except where the vertical spaces increase or are greater on the roof sides. Then the line is broken. Therefore, it is only necessary to have a dimension at the top and bottom row for the spaces coming within the straight line and as shown by the 11 3/16-inch top, 19 13/16 inch bottom, and 32⅝-inch vertical dimensions for this boiler. All scaled dimensions for radial stays and staybolts should be to the nearest 1/16 inch. Refer to "Features of Boiler Design" for nomenclature of "radial stays" and "staybolts."

SEAM RIVETS

The roof-to-backhead-seam rivets are spaced on the flat sheet according to the pitch given on the boiler drawing. Scale the distance between the top center and bottom side rivet, using a flexible scale or stepping off with the dividers on a straight line and then scaling. Divide this dimension by the pitch.

The rivet spacing in the top half of the front seam is calculated at the line where shearing occurs, that is, where the underside or water side of the roof sheet and the outside of the ring course come together. The size and pitch of the rivets are given on the boiler drawing and the number of spaces should be divisible by four to come on the three centers of the ring. The number of spaces nearest to the given pitch, i.e., divisible by four should be chosen. For the rivets in the bottom half of the sheet, scale the distance from the ring center to the bottom rivet and divide by the same rivet pitch. Rivets in the bottom half of the roof sheet for narrow firebox boilers, have the same pitch down to the throat seam as the top half of the sheet, and then are scaled from this point to the bottom rivet. Show a center line for all the rivets in these seams. The firebox ring rivets on the roof sheet are spaced the same as those on the crown sheet.

Large circles with the same diameter as the sleeve are shown for the flexible staybolts and expansion stays and small circles of the same diameter as the staybolt for the rigid staybolts. A note on the drawing—Fig. 8—refers to the size and kind of staybolts.

This completes the roof and side sheet shop drawing up to the point of checking.

CHECKING THE CROWN AND ROOF SHEET DRAWINGS

The work of inspecting the correctness of a drawing is termed "checking" the drawing. In order for a drawing to

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